

Missense variants identified in individuals with KURIS/NDD and benign/likely-benign ClinVar missense variants, annotated with the affected protein domain, the SGE functional score and GMM class, ClinVar classification and review status, and associated clinical information. Each variant's clinical occurrences are listed on a labelled second row.

Gene: HGNC ID	Category	HGVS g.	HGVS p.	HGVS c.	SGE score	Domain	GMM class	ClinVar (agg. germline)	ClinVar VCV	Stars
BAP1; HGNC:950	KURIS/NDD	NC_000003.12:g.52409585C>T	NP_004647.1:p.Glu31Lys	NM_004656.4:c.91G>A	-0.201	UCH	Abnormal	Conflicting classifications of pathogenicity	VCV001173078	1
ClinVar KURIS lab entry (SCV): SCV001738369: P, Cond: Neurodevelopmental disorder, MedGenet, 2021-06-22 [de novo; affected: yes] Küry et al. 2022: F3 (Sex: female; Age_at_last_investigation: 14 years; Growth_failure: +; Developmental_delay_or_ID: +; mild; Behavioral_anomalies: +; ADHD; Cardiac_anomalies: +; Eye_anomalies: +; Feeding_difficulties: +; Facial_dysmorphism: +; Other_recurrent_signs: frequent otitis)										
BAP1; HGNC:950	KURIS/NDD	NC_000003.12:g.52408061C>G	NP_004647.1:p.Cys91Ser	NM_004656.4:c.272G>C	-0.177	UCH	Abnormal	Pathogenic	VCV001173081	1
ClinVar KURIS lab entry (SCV): SCV001738372: P, Cond: Neurodevelopmental disorder, MedGenet, 2021-06-22 [de novo; affected: yes] Küry et al. 2022: F5 (Sex: female; Age_at_last_investigation: 11 years 5 months; Growth_failure: +; Developmental_delay_or_ID: +; severe; Seizures: +; Urogenital_kidney_anomalies: +; Feet: +; Facial_dysmorphism: +; Other_recurrent_signs: hypertrichosis)										
BAP1; HGNC:950	KURIS/NDD	NC_000003.12:g.52407247G>C	NP_004647.1:p.His169Gln	NM_004656.4:c.507C>G	-0.169	UCH	Abnormal	Likely pathogenic	VCV003132975	1
ClinVar KURIS lab entry (SCV): SCV004911795: LP, Ambry Genetics, 2023-11-20 [This alteration results from a C to G substitution at nucleotide position 507, causing the histidine at amino acid position 169 to be replaced by a glutamine for BAP1-related neurodevelopmental disorder; however, its clinical significance for BAP1-related tumor predisposition syndrome is uncertain. This variant was not reported in population-based cohorts in gnomAD. Another alteration at the same codon, c.506A>G (p.H169R), has been detected as de novo in two individuals with a syndromic form of intellectual disability and/or developmental delay (Kury, 2022).]										
BAP1; HGNC:950	KURIS/NDD	NC_000003.12:g.52409851T>G	NP_004647.1:p.Ser10Arg	NM_004656.4:c.28A>C	-0.162	UCH	Abnormal			
DECIPHER: Patient 534278 (Pathogenic; 46XY; De novo (parentage confirmed)) [Bifid uvula; Broad eyebrow; Disproportionate short stature; Intellectual disability; Osteoporosis]										
BAP1; HGNC:950	KURIS/NDD	NC_000003.12:g.52408062A>C	NP_004647.1:p.Cys91Gly	NM_004656.4:c.271T>G	-0.161	UCH	Abnormal	Uncertain significance	VCV001173082	1
ClinVar KURIS lab entry (SCV): SCV001738373: P, Cond: Neurodevelopmental disorder, MedGenet, 2021-06-22 [de novo; affected: yes] Küry et al. 2022: F6 (Sex: female; Age_at_last_investigation: 15 years; Growth_failure: +; Developmental_delay_or_ID: +; severe; Behavioral_anomalies: +; ASD; Eye_anomalies: +; Hands: +; Feet: +; Feeding_difficulties: +; Facial_dysmorphism: +; Other_recurrent_signs: hypertrichosis)										
BAP1; HGNC:950	KURIS/NDD	NC_000003.12:g.52407248T>C	NP_004647.1:p.His169Arg	NM_004656.4:c.506A>G	-0.137	UCH	Abnormal	Conflicting classifications of pathogenicity	VCV001173083	1
ClinVar KURIS lab entry (SCV): SCV001738375: P, Cond: Neurodevelopmental disorder, MedGenet, 2021-06-22 [de novo; affected: yes]; SCV002098944: P, Cond: KURY-ISIDOR SYNDROME, OMIM, 2022-02-24 [Patient 9 was a 16-year-old boy with mild developmental delay, behavioral problems noted as sensitivity, seizures, distal skeletal anomalies, and patchy alopecia. He also carried biallelic variants in the NCSTN gene (605254). Patient 10 was an 8-year-old girl with mild global developmental delay, poor speech, a history of seizures, behavioral abnormalities, and mild dysmorphic features] Küry et al. 2022: F9 (Sex: male; Age_at_last_investigation: 16 years; Growth_failure: +; Developmental_delay_or_ID: +; Seizures: +; Behavioral_anomalies: +; Hands: +; Feet: +; Facial_dysmorphism: +; Other_recurrent_signs: hearing loss); F10 (Sex: female; Age_at_last_investigation: 8 years 4 months; Growth_failure: +; Developmental_delay_or_ID: +; Seizures: +; Behavioral_anomalies: +; Cardiac_anomalies: +; VSD; Eye_anomalies: +; Urogenital_kidney_anomalies: +; Feeding_difficulties: +; Facial_dysmorphism: +)										
BAP1; HGNC:950	KURIS/NDD	NC_000003.12:g.52409845G>T	NP_004647.1:p.Pro12Thr	NM_004656.4:c.34C>A	-0.137	UCH	Abnormal	Pathogenic	VCV001064792	0
ClinVar KURIS lab entry (SCV): SCV001572228: P, Cond: Neurodevelopmental disorder, MedGenet, 2021-06-22 [de novo; affected: yes]; SCV002098941: P, Cond: KURY-ISIDOR SYNDROME, OMIM, 2022-02-24 Küry et al. 2022: F1 (Sex: female; Age_at_last_investigation: 10 years; Developmental_delay_or_ID: +; Seizures: +; Behavioral_anomalies: +; ADHD; Facial_dysmorphism: +)										
BAP1; HGNC:950	KURIS/NDD	NC_000003.12:g.52408062A>G	NP_004647.1:p.Cys91Arg	NM_004656.4:c.271T>C	-0.135	UCH	Abnormal	Conflicting classifications of pathogenicity	VCV000988095	1
ClinVar KURIS lab entry (SCV): SCV001738374: P, Cond: Neurodevelopmental disorder, MedGenet, 2021-06-22 [de novo; affected: yes]; SCV002098943: P, Cond: KURY-ISIDOR SYNDROME, OMIM, 2022-02-24; SCV003921047: P, UKL, 2023-02-20 [de novo; affected: yes; ACMG: PP3; This variant was identified as de novo (maternity and paternity confirmed). Criteria applied: PS2_VSTR, PM5_STR, PS4_MOD, PM2_SUP, PP3]; SCV001449490: VUS, Cond: BAP1-associated neurodevelopmental disorder, UDN, 2020-06-12 [de novo; affected: yes; clinical features: Square face, Piliomatixoma, Nystagmus, Hypertrichosis, Severe Myopia, Growth delay, Global developmental delay, Generalized hypotonia, Fused teeth, Focal seizures, Exudative retinopathy, Dysphagia, Broad-based gait, Brachycephaly, Anteverted nares, Abnormality of the pinna]; SCV004801702: P, Cond: see cases, IHG, 2023-12-13 [affected: yes; clinical features: Global developmental delay, Microcephaly, Poor speech, Infantile muscular hypotonia, Single transverse palmar crease; ACMG: BP1;PM1;PM2;PM5;PP3;PS4; ACMG categories: PS4,PM1,PM2,PM5,PP3,BP1] Küry et al. 2022: F7 (Sex: male; Age_at_last_investigation: 1 years 11 months; Growth_failure: +; Developmental_delay_or_ID: +; Facial_dysmorphism: +; Other_recurrent_signs: recurrent otitis); F8 (Sex: female; Age_at_last_investigation: 4 years; Growth_failure: +; Developmental_delay_or_ID: +; Seizures: +; Behavioral_anomalies: +; Eye_anomalies: +; Feeding_difficulties: +; Facial_dysmorphism: +)										
BAP1; HGNC:950	KURIS/NDD	NC_000003.12:g.52408583A>G	NP_004647.1:p.Leu49Pro	NM_004656.4:c.146T>C	-0.135	UCH	Abnormal	Likely pathogenic	VCV001173079	1
ClinVar KURIS lab entry (SCV): SCV001738370: P, Cond: Neurodevelopmental disorder, MedGenet, 2021-06-22 [de novo; affected: yes] Küry et al. 2022: F4 (Sex: male; Age_at_last_investigation: 6 years; Growth_failure: +; Developmental_delay_or_ID: +; Behavioral_anomalies: +; ASD; Hands: +; Facial_dysmorphism: +; Other_recurrent_signs: hypertrichosis)										
BAP1; HGNC:950	KURIS/NDD	NC_000003.12:g.52408553C>G	NP_004647.1:p.Arg59Pro	NM_004656.4:c.176G>C	-0.102	UCH	Abnormal	Pathogenic	VCV001173080	1
ClinVar KURIS lab entry (SCV): SCV001738371: P, Cond: Neurodevelopmental disorder, MedGenet, 2021-06-22 [de novo; affected: yes] DECIPHER: Patient 550703 (Likely pathogenic; 46XX; De novo (parentage confirmed))										
BAP1; HGNC:950	KURIS/NDD	NC_000003.12:g.52409845G>C	NP_004647.1:p.Pro12Ala	NM_004656.4:c.34C>G	-0.078	UCH	Abnormal	Likely pathogenic	VCV001173077	1
ClinVar KURIS lab entry (SCV): SCV001738368: P, Cond: Neurodevelopmental disorder, MedGenet, 2021-06-22 [de novo; affected: yes]; SCV002098942: P, Cond: KURY-ISIDOR SYNDROME, OMIM, 2022-02-24 Küry et										

Gene; HGNC ID	Category	HGVS g.	HGVS p.	HGVS c.	SGE score	Domain	GMM class	ClinVar (agg. germline)	ClinVar VCV	Stars
<i>BAP1</i> ; HGNC:950	KURIS/NDD	NC_000003.12:g.52408550C>T	NP_004647.1:p.Arg60Gln	NM_004656.4:c.179G>A	-0.018	UCH	Normal	Conflicting classifications of pathogenicity	VCV000489626	1
ClinVar KURIS lab entry (SCV): SCV006277912: LP, IMGAG, 2025-07-10 [affected: yes; clinical features: Low frustration tolerance, Delayed speech and language development, Hypertrichosis, Global developmental delay, Motor delay, Small for gestational age, Short stature, Attention deficit hyperactivity disorder, Feeding difficulties in infancy, Neurodevelopmental delay, Difficulty walking on uneven ground, Hand flapping]										
<i>BAP1</i> ; HGNC:950	KURIS/NDD	NC_000003.12:g.52402325C>T	NP_004647.1:p.Arg718Gln	NM_004656.4:c.2153G>A	0.003	BRCA1-binding	Normal	Uncertain significance	VCV001173084	2
ClinVar KURIS lab entry (SCV): SCV001738376: P, Cond: Neurodevelopmental disorder, MedGenet, 2021-06-22 [de novo; affected: yes] Küry et al. 2022: F11 (Sex: female; Age_at_last_investigation: 12 years 1 month; Growth_failure: +; Developmental_delay_or_ID: +; Seizures: +; Behavioral_anomalies: +; ADHD)										
<i>BAP1</i> ; HGNC:950	KURIS/NDD	NC_000003.12:g.52408553C>T	NP_004647.1:p.Arg59Gln	NM_004656.4:c.176G>A	0.007	UCH	Normal	Conflicting classifications of pathogenicity	VCV001320722	1
ClinVar KURIS lab entry (SCV): SCV006586709: LP, IMGAG, 2025-10-24 [de novo; affected: yes; clinical features: Telecanthus, Myopia, Atypical behavior, Intellectual disability, Global developmental delay]										
<i>BAP1</i> ; HGNC:950	B/LB	NC_000003.12:g.52403199C>T	NP_004647.1:p.Arg610Lys	NM_004656.4:c.1829G>A	-0.016	BRCA1-binding	Normal	Likely benign	VCV003224444	2
<i>BAP1</i> ; HGNC:950	B/LB	NC_000003.12:g.52403242T>C	NP_004647.1:p.Ser596Gly	NM_004656.4:c.1786A>G	-0.007	BRCA1-binding	Normal	Benign	VCV000133664	2
<i>BAP1</i> ; HGNC:950	B/LB	NC_000003.12:g.52405827T>G	NP_004647.1:p.Asn290Thr	NM_004656.4:c.869A>C	-0.005	Other	Normal	Likely benign	VCV003477925	1
<i>BAP1</i> ; HGNC:950	B/LB	NC_000003.12:g.52405279G>T	NP_004647.1:p.Ala316Glu	NM_004656.4:c.947C>A	-0.004	Other	Normal	Likely benign	VCV002565171	1
<i>BAP1</i> ; HGNC:950	B/LB	NC_000003.12:g.52404485C>G	NP_004647.1:p.Glu406Asp	NM_004656.4:c.1218G>C	-0.002	Other	Normal	Likely benign	VCV001752271	1
<i>BAP1</i> ; HGNC:950	B/LB	NC_000003.12:g.52403274G>A	NP_004647.1:p.Ser585Phe	NM_004656.4:c.1754C>T	-0.002	Other	Normal	Likely benign	VCV003477483	1
<i>BAP1</i> ; HGNC:950	B/LB	NC_000003.12:g.52403292C>T	NP_004647.1:p.Gly579Glu	NM_004656.4:c.1736G>A	-0.001	Other	Normal	Benign	VCV002450425	1
<i>BAP1</i> ; HGNC:950	B/LB	NC_000003.12:g.52403718A>G	NP_004647.1:p.Val476Ala	NM_004656.4:c.1427T>C	-0.001	Other	Normal	Benign/Likely benign	VCV000240048	2
<i>BAP1</i> ; HGNC:950	B/LB	NC_000003.12:g.52402827G>C	NP_004647.1:p.Asn645Lys	NM_004656.4:c.1935C>G	-0.001	BRCA1-binding	Normal	Benign	VCV004194475	1
<i>BAP1</i> ; HGNC:950	B/LB	NC_000003.12:g.52405812A>G	NP_004647.1:p.Val295Ala	NM_004656.4:c.884T>C	-0.001	Other	Normal	Benign	VCV003477649	1
<i>BAP1</i> ; HGNC:950	B/LB	NC_000003.12:g.52404549C>G	NP_004647.1:p.Arg385Pro	NM_004656.4:c.1154G>C	-0.001	Other	Normal	Likely benign	VCV000818436	1
<i>BAP1</i> ; HGNC:950	B/LB	NC_000003.12:g.52403235G>A	NP_004647.1:p.Pro598Leu	NM_004656.4:c.1793C>T	0.001	BRCA1-binding	Normal	Benign	VCV003818386	1
<i>BAP1</i> ; HGNC:950	B/LB	NC_000003.12:g.52408590T>C	NP_004647.1:p.Ile47Val	NM_004656.4:c.139A>G	0.001	UCH	Normal	Benign	VCV001771715	1
<i>BAP1</i> ; HGNC:950	B/LB	NC_000003.12:g.52403877G>T	NP_004647.1:p.Thr423Lys	NM_004656.4:c.1268C>A	0.002	Other	Normal	Benign/Likely benign	VCV000240045	2
<i>BAP1</i> ; HGNC:950	B/LB	NC_000003.12:g.52402828T>C	NP_004647.1:p.Asn645Ser	NM_004656.4:c.1934A>G	0.003	BRCA1-binding	Normal	Likely benign	VCV003260366	1
<i>BAP1</i> ; HGNC:950	B/LB	NC_000003.12:g.52403162C>G	NP_004647.1:p.Leu622Phe	NM_004656.4:c.1866G>C	0.003	BRCA1-binding	Normal	Benign	VCV004620225	1
<i>BAP1</i> ; HGNC:950	B/LB	NC_000003.12:g.52405795C>A	NP_004647.1:p.Ala301Ser	NM_004656.4:c.901G>T	0.004	Other	Normal	Benign	VCV004620599	1
<i>BAP1</i> ; HGNC:950	B/LB	NC_000003.12:g.52407418C>T	NP_004647.1:p.Ala140Thr	NM_004656.4:c.418G>A	0.004	UCH	Normal	Likely benign	VCV004194127	1
<i>BAP1</i> ; HGNC:950	B/LB	NC_000003.12:g.52405791G>A	NP_004647.1:p.Pro302Leu	NM_004656.4:c.905C>T	0.004	Other	Normal	Benign/Likely benign	VCV000133661	2
<i>BAP1</i> ; HGNC:950	B/LB	NC_000003.12:g.52405827T>C	NP_004647.1:p.Asn290Ser	NM_004656.4:c.869A>G	0.005	Other	Normal	Benign/Likely benign	VCV000472716	2
<i>BAP1</i> ; HGNC:950	B/LB	NC_000003.12:g.52408493T>C	NP_004647.1:p.Asn79Ser	NM_004656.4:c.236A>G	0.005	UCH	Normal	Likely benign	VCV003224460	1
<i>BAP1</i> ; HGNC:950	B/LB	NC_000003.12:g.52409555C>T	NP_004647.1:p.Gly41Ser	NM_004656.4:c.121G>A	0.006	UCH	Normal	Benign/Likely benign	VCV000133667	2
<i>BAP1</i> ; HGNC:950	B/LB	NC_000003.12:g.52404520A>C	NP_004647.1:p.Ser395Ala	NM_004656.4:c.1183T>G	0.007	Other	Normal	Benign	VCV003986661	1
<i>BAP1</i> ; HGNC:950	B/LB	NC_000003.12:g.52403190G>A	NP_004647.1:p.Thr613Met	NM_004656.4:c.1838C>T	0.009	BRCA1-binding	Normal	Benign/Likely benign	VCV000133665	2
<i>BAP1</i> ; HGNC:950	B/LB	NC_000003.12:g.52407990T>G	NP_004647.1:p.Met115Leu	NM_004656.4:c.343A>C	0.014	UCH	Normal	Benign	VCV004620701	1
<i>BAP1</i> ; HGNC:950	B/LB	NC_000003.12:g.52403646C>T	NP_004647.1:p.Gly500Asp	NM_004656.4:c.1499G>A	0.014	Other	Normal	Likely benign	VCV004620450	1
<i>BAP1</i> ; HGNC:950	B/LB	NC_000003.12:g.52409574G>C	NP_004647.1:p.Asp34Glu	NM_004656.4:c.102C>G	0.014	UCH	Normal	Likely benign	VCV003986856	1
<i>BAP1</i> ; HGNC:950	B/LB	NC_000003.12:g.52403799G>A	NP_004647.1:p.Ala449Val	NM_004656.4:c.1346C>T	0.016	Other	Normal	Benign	VCV003477759	1